

Morphology ₁

- Introduction
- Morphology
- Morphological Analysis (MA)
- Using FS techniques in MA
- Automatic learning of the morphology of a language

Morphology ²

- Morphology
 - Structure of a word as a composition of morphemes
 - Related to word formation rules
 - Functions
 - Inflection
 - Derivation
 - Composition
- Result of morphologic analysis
 - Morphosyntactic categorization (POS)
 - e.g. Parole tagset (VMIP1S0), more than 150 categories for Spanish
 - e.g. Penn Treebank tagset (VBD), about 30 categories for English
 - Morphological features
 - Number, case, gender, lexical functions

Morphology ³

- Morphologic analysis
 - Decompose a word into a concatenation of morphemes
 - Usually some of the morphemes contain the meaning
 - One (root or stem) in flexion and derivation
 - More than one in composition
 - The other (affixes) provide morphological features
- Problems
 - Phonological alterations in morpheme concatenation
 - Morphotactics
 - Which morphemes can be concatenated with which others

- Problems
 - Affixes
 - Suffixes, prefixes, infixes, interfixes
 - Inflectional affixes $\neq$ derivational affixes
 - Derivation implies sometimes a semantic change not always predictable
 - Meaning extensions
 - Lexical rules
 - A derivational suffix can be followed by an inflectional one
 - love $\Rightarrow$ lover $\Rightarrow$ lovers
 - Inflection does not change POS, sometimes derivation does
 - Inflection affects other words in the sentence
 - agreement

Morphology ⁵

- Morphotactics
 - Word formation rules
 - Valid combinations between morphemes
 - Simple concatenation
 - Complex models root/pattern
 - Language dependency regularity
- Phonological alterations (Morphophonology)
 - Changes when concatenating morphemes
 - Source: Phonology, morphology, orthography
 - variable in number and complexity
 - e.g. vocalic harmony

Morphology ⁶

Morphemes

- 1 morpheme:
Evitar (verb to avoid)
- 2 morphemes:
 - evitable = evitar + able (adj: can be avoided)
- 3 morphemes:
 - inevitable = in + evitar + able
(adj: cannot be avoided)
- 4 morphemes:
 - inevitabilidad = in + evitar + able + idad
(noun: cannot be avoided)

Morphology ⁷

Inflectional Morphology

- number
 - house houses
 - cheval chevaux
 - casa casas
- verbal form
 - walk walkes walked walking
 - amo amas aman ...
- gender
 - niño niña

Morphology ⁸

Derivational Morphology

- Form
 - Without change barcelonés
 - Prefix inevitable
 - Suffix importantísimo
- Source
 - verb => adjective tardar => tardío
 - verb => noun sufrir => sufrimiento
 - noun => noun actor => actorazo
 - noun => adjective atleta => atlético
 - adjective => adjective rojo => rojizo
 - adjective => adverb alegre => alegremente

Morphological Analysis ¹

Types of morphological analyzers

Formaries

- Dictionaries of word forms
 - + efficiency
 - + Languages with few variants (e.g. English)
 - + extensibility
 - + Possibility of building and maintenance from a morphological generator
 - Languages with high flexive variation
 - derivation, composition
- FS techniques
 - FSA
 - 1 level analyzers
 - FST
 - > 1 level analyzers

Morphological Analysis ₂

Morphological analyzers of two levels

- General model for languages with morpheme concatenation
- Independence between lingware and analyzer
- Valid for analysis and generation
- Distinction between lexical and superficial levels
- Parallel rules for morphophonology
- Simple implementation

Morphological Analysis ₃

- Morphological rules
 - Define the relations between characters (surface) and morphemes and map strings of characters and the morphemic structure of the word.
- Spelling rules
 - Perform at the level of the letters forming the word. Can be used to define the valid phonological alterations.
- Ritchie, Pulman, Black, Russell, 1987

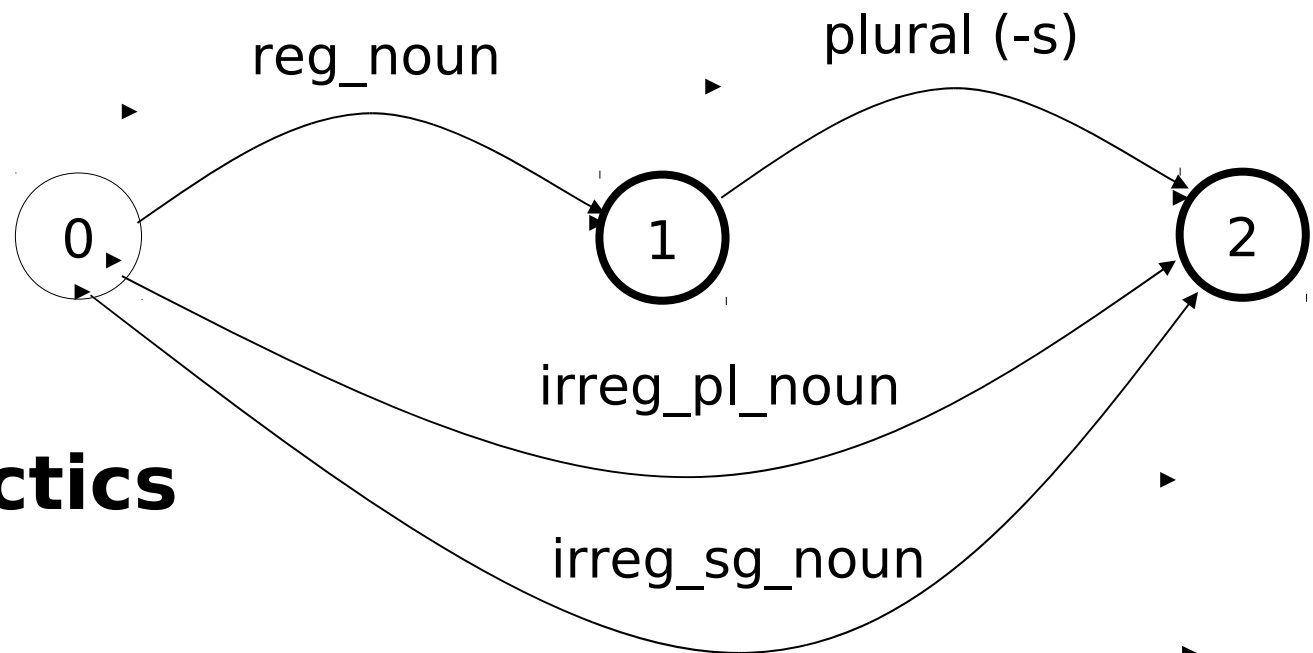
Morphological Analysis ₄

- input:
 - form
- output
 - lemma + morphological features

Input	Output
cat	cat + N + sg
cats	cat + N + pl
cities	city + N + pl
merging	merge + V + pres_part
caught	(catch + V + past) or (catch + V + past_part)

Morphological Analysis ₅

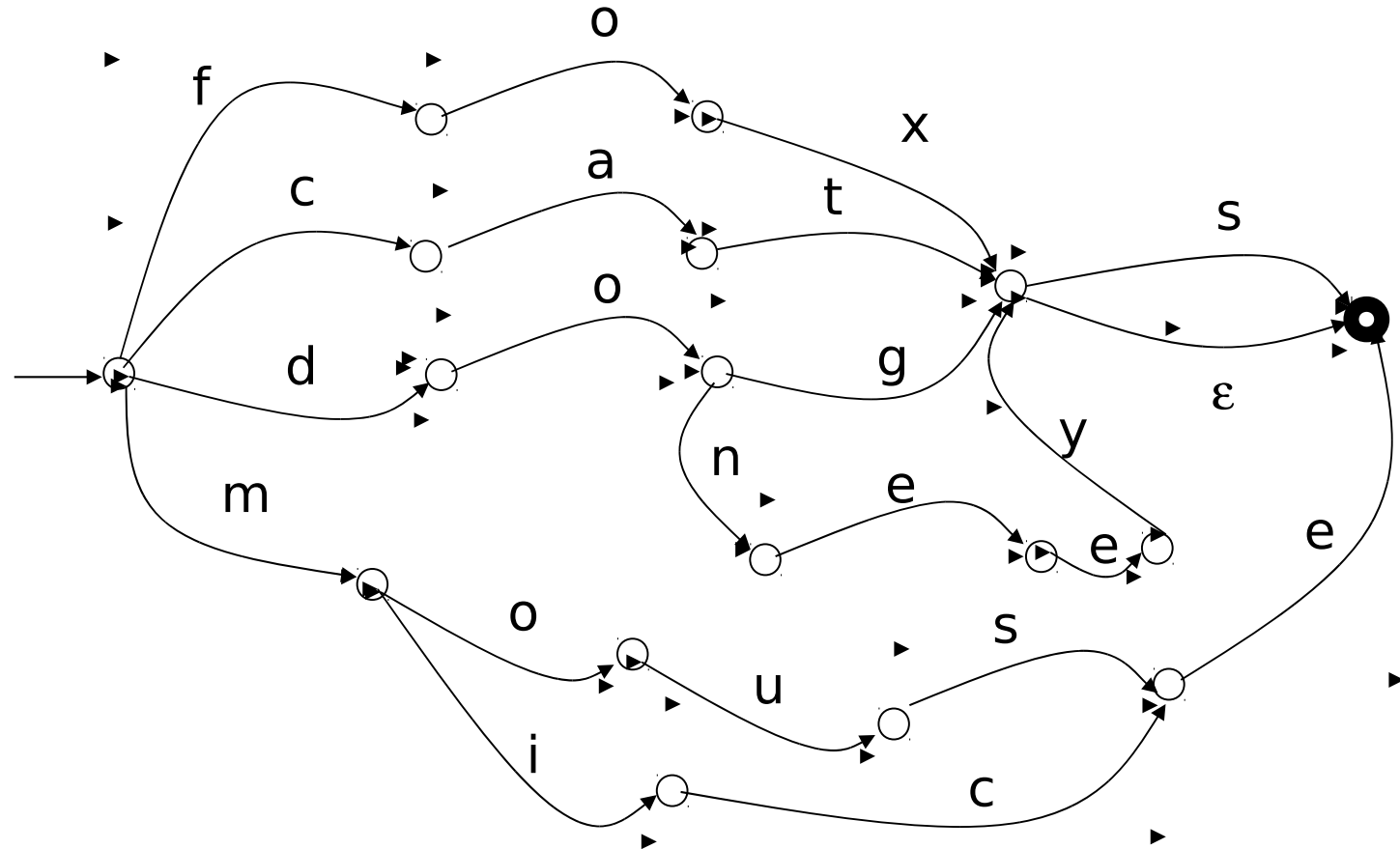
reg_noun	irreg_pl_noun	irreg_sg_noun	plural
fox cat dog	sheep mice	sheep mouse	-s



Morphotactics

Morphological Analysis ₆

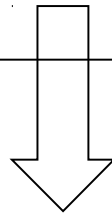
fog
cat
dog
donkey
mouse
mice



Letter Transducers

Morphological Analysis ⁷

upper level	lexic	cat + N	cat
+ N + pl			
lower level	surface	cat	
cats			



c:c	a:a	t:t	+N:ε	+pl:s
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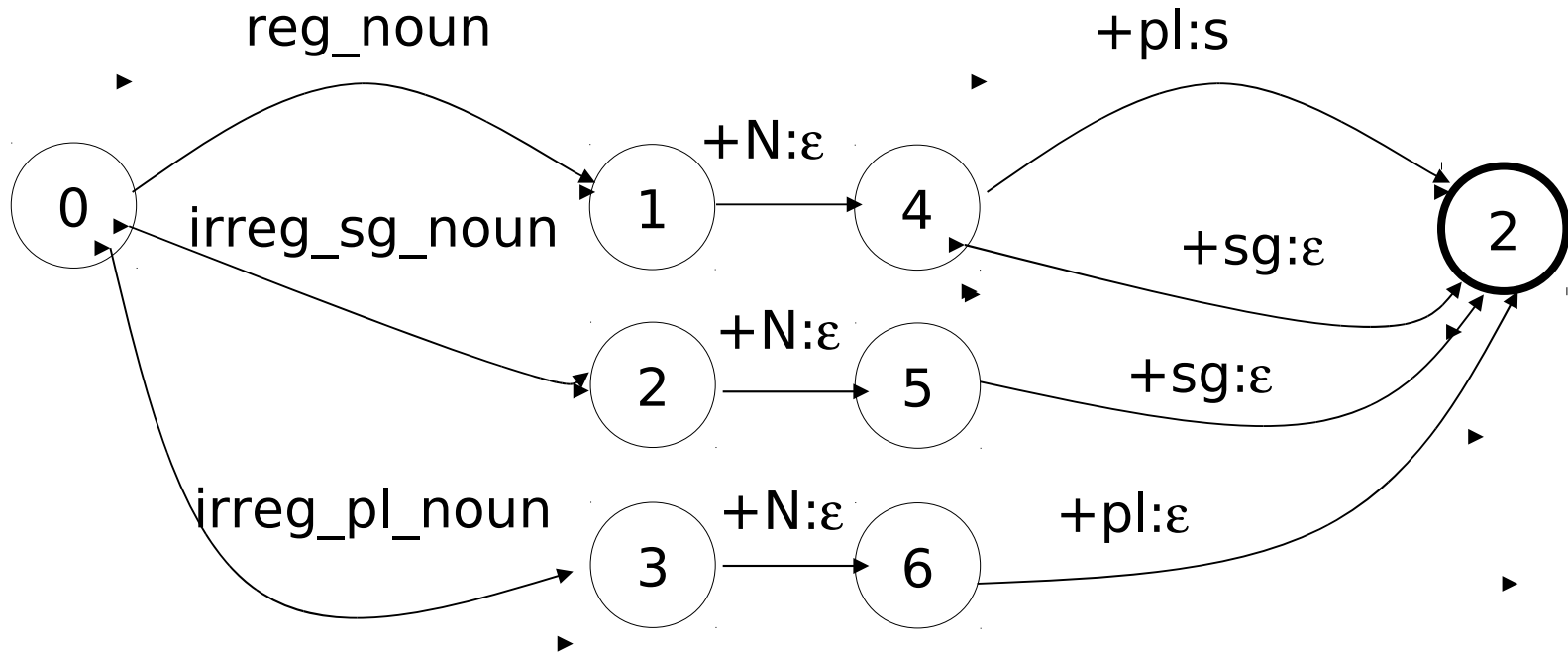
Morphological Analysis ₈

Using FST

- As a recognizer
 - From a pair of input strings (one lexical and the other superficial) determines if one is transduction of the other
- As a generator
 - Generates pairs of strings
- As a translator
 - From a superficial string generates its lexical translation

Morphological Analysis ⁹

reg_noun	irreg_pl_noun	irreg_sg_noun	plural
fox	sheep	sheep	s
cat	m o:i u:ε ce	mouse	
dog	g o:e o:e se	goose	



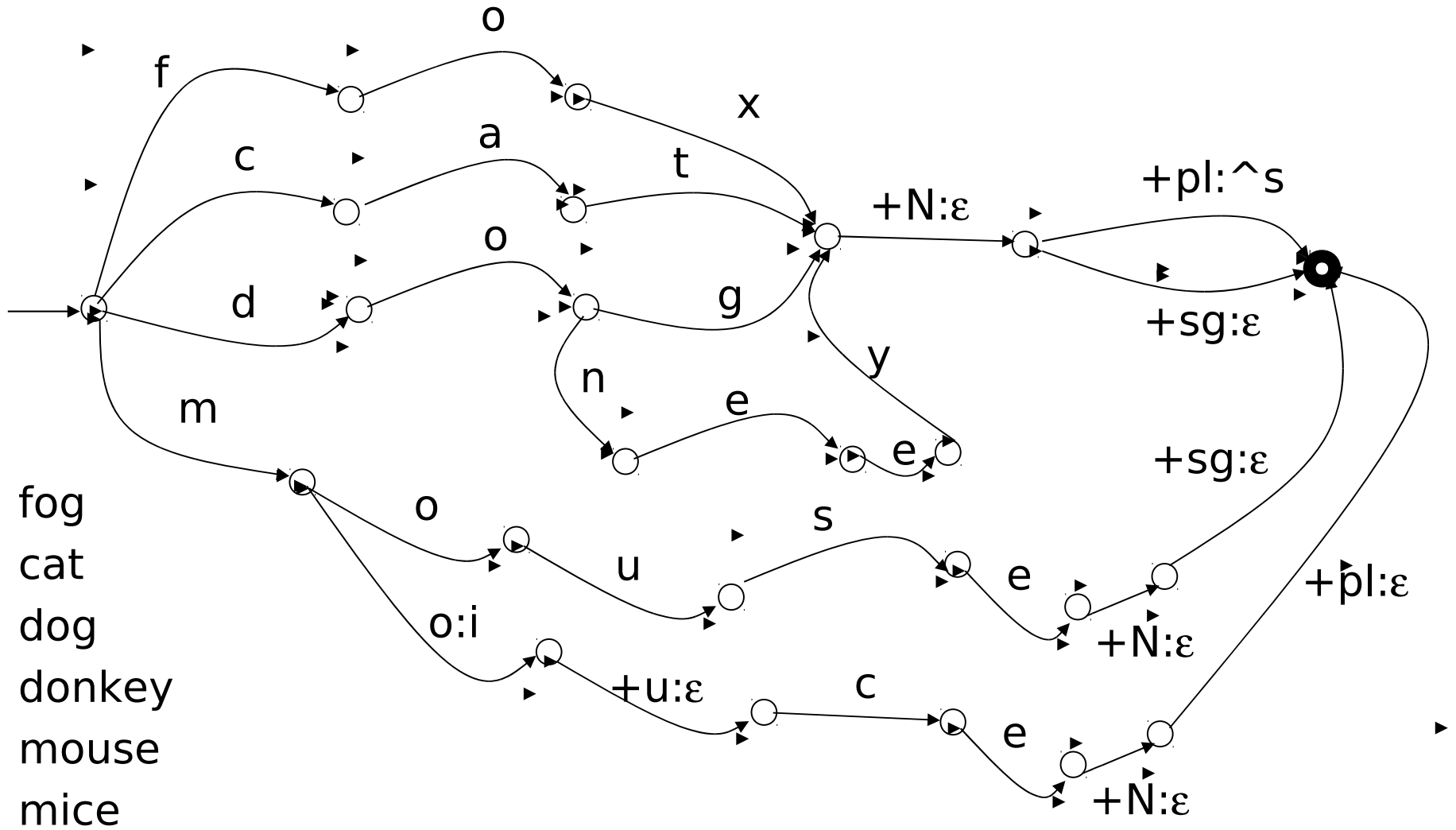
Morphological Analysis 10

morphotactics

spelling rules

lexical level	f	o	x	+N	+pl
intermediate level	f	o	x	^	s
superficial level		f	o	x	e
s					

Morphological Analysis 11



Morphological Analysis ¹²

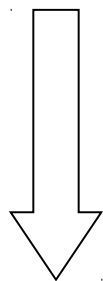
Spelling rules

name	description	example
consonant doubling beg/begging	single letter consonant doubled before -ing/-ed	
e deletion	silent e dropped before -ing/-ed	
make/making e insertion	e added after -s,-z,-x,-ch,-sh	
y replacement	before -s -y changes to -ie before -s, to	watch/watches
k insertion	i before -ed verbs ending with vowel +c add -k	try/tries panic/panicked

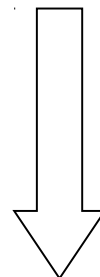
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Spelling rules: e-insertion

$$\varepsilon : e \iff [xsz]^\wedge : \varepsilon \quad \underline{\hspace{1cm}} \quad s\#$$



decomposition



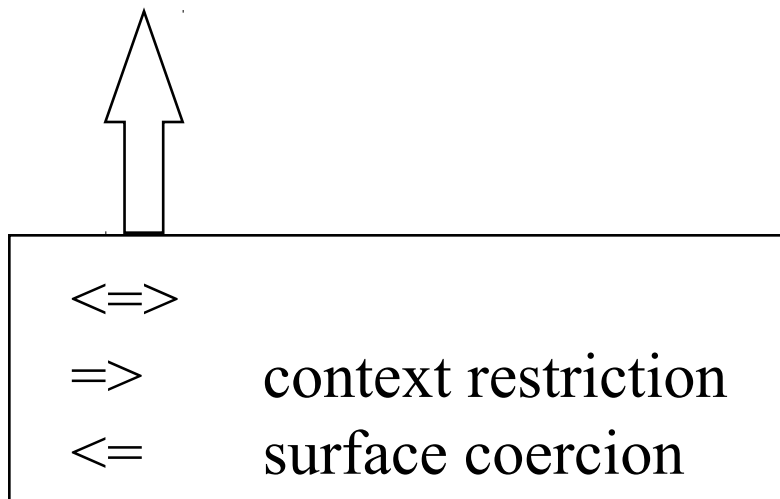
$$\varepsilon : e \Rightarrow [xsz]^\wedge : \varepsilon \quad s\#$$

$$\varepsilon:\varepsilon \ / \Leftarrow [XSZ]^\wedge:\varepsilon \quad S\#$$

Morphological Analysis ¹⁴

epenthesis

+ : e $\langle \Rightarrow \rangle$ { < {s:s c:c} h:h > s:s x:x z:z } $\dashv$ s:s



context

C: {...}
V: {a,e,i,o,u,y}
C2: {...}
=: whatever

example: $\frac{\text{box} \quad + \quad \text{s}}{\text{box} \quad \text{e} \quad \text{s}}$

e-deletion

$e : 0 \rightarrow \begin{matrix} <=> \\ \text{or} \\ \text{or} \\ \text{or} \\ \text{or} \end{matrix} \begin{matrix} = :C2 \\ <C:C \ V:V> \\ <c:c \ g:g> \\ l:0 \\ c:c \end{matrix} \begin{matrix} --- \\ --- \\ --- \\ --- \\ --- \end{matrix} \begin{matrix} <+:0 \ V:=> \\ <+:0 \ e:e> \\ <+:0 \ \{e:e \ i:i\}> \\ +:0 \\ <+:0 \ a:0 \ t:t \ b:b> \end{matrix}$

mov	e	+	ed
mov			ed

agre	e	+	ed
agre			ed

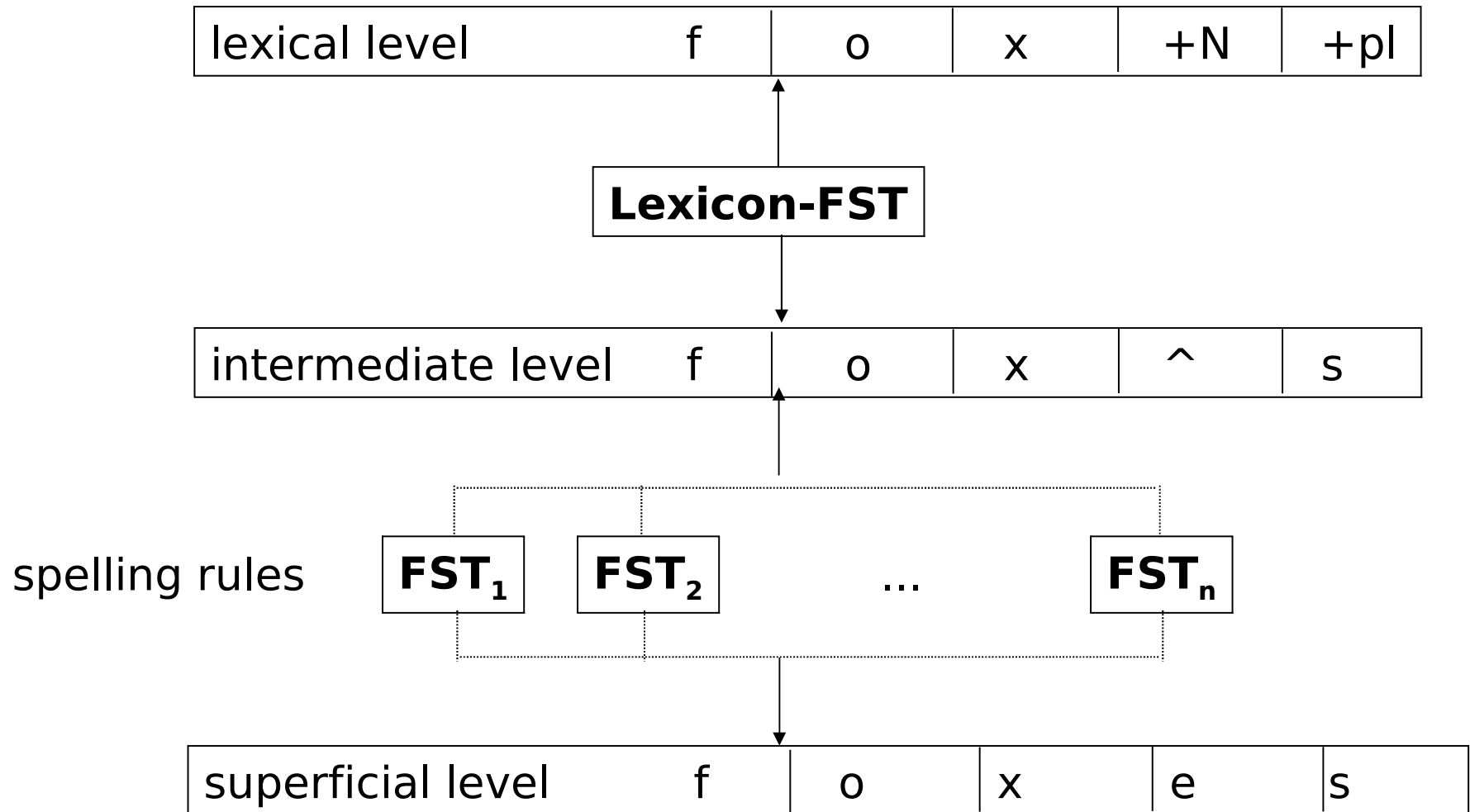
a-deletion

$a : 0 \quad \Leftrightarrow \quad \langle c:c \ e:0 \ +:0 \rangle \quad \text{---} \quad t:t$

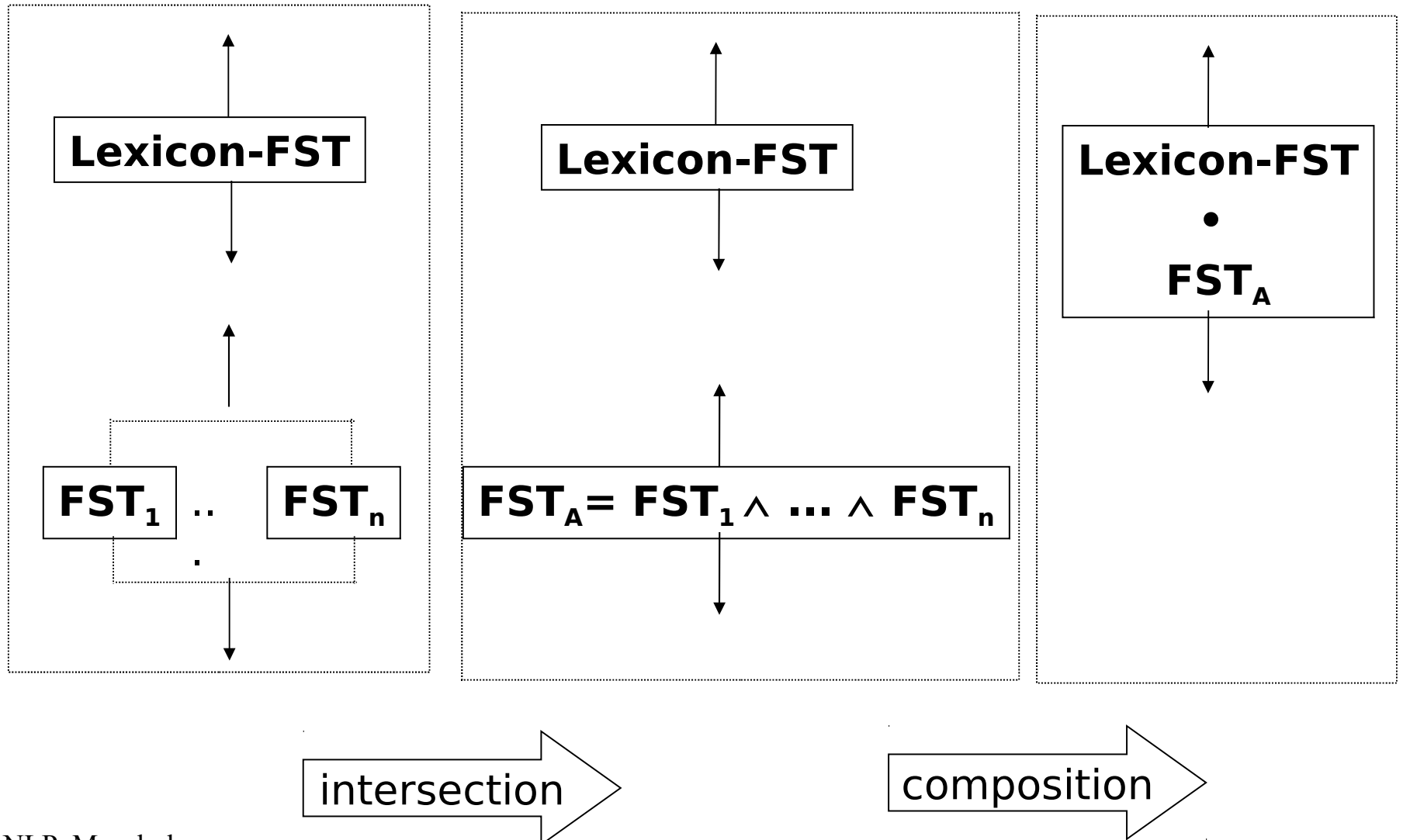
redu	c	e	+	a	t	ion
redu	c				t	ion

... left context focus right context ...

Morphological Analysis ¹⁷



Morphological Analysis ¹⁸



- Problem
 - Paradigm stem + affixes
 - Obtaining the stems
 - Classification of stems into models
 - Learning part of the morphology (e.g. derivational)
- Two approaches
 - No previous morphologic knowledge is available
 - Goldsmith, 2001
 - Brent, 1999
 - Snover, Brent, 2001, 2002
 - Morphologic knowledge can be used
 - Oliver et al, 2002

Automatic morphology learning ₂

- Automatic morphological analysis
 - Identification of borders between morphemes
 - Zellig Harris
 - {prefix, suffix} conditional entropy
 - bigrams and trigrams with high probability of forming a morpheme
 - Learning of patterns or rules of mapping between pairs of words
 - Global approach (top-down)
 - Golsdmith, Brent, de Marcken

- Goldsmith's system based on MDL (Minimum Description Length)
 - Initial Partition: word $\rightarrow$ stem + suffix
 - split-all-words
 - A good candidate to {stem, suffix} splitting in a word has to be a good candidate in many other words
 - MI (mutual information) strategy
 - Faster convergence
 - Learning Signatures
 - {signatures, stem, suffixes}
 - MDL

- Semi-automatic morphological analysis
 - Oliver, 2004
 - Starts with a set of manually written morphological rules
 - TL:TF:Desc
 - lemma ending
 - form ending
 - POS
 - Lists of non flexive classes , closed classes and irregular words
 - Corpora
 - Serbo-Croatian 9 Mw
 - Russian 16 Mw